

Basics 12: Maxwell Equations

II - Electromagnetism foundations

Local and integral laws, conservation of charge, electromagnetic waves, and plasma limits

Textbook draft, version 1.0. Maxwell's equations organize classical electromagnetism into four coupled field laws. This chapter derives their integral forms, obtains charge conservation and vacuum electromagnetic waves, and then identifies the low-frequency approximations used in magnetohydrodynamics. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- State Maxwell’s equations in differential and integral form.
- Derive the continuity equation for electric charge.
- Derive the electromagnetic wave equation in vacuum.
- Explain the displacement-current term.
- Distinguish exact field constraints from model-dependent plasma approximations.
- Identify numerical consequences of divergence constraints.

2 Prerequisites and reading strategy

- Basics 2, 3, and 11.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 The four differential equations

In SI units,

$$\nabla \cdot \mathbf{E} = \frac{\rho_q}{\epsilon_0}, \quad (3.1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (3.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (3.3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (3.4)$$

Gauss's electric law relates charge to electric flux. Gauss's magnetic law states that the net magnetic flux through a closed surface vanishes. Faraday's law describes induction. Ampere-Maxwell law connects magnetic circulation to conduction current and changing electric flux.

4 Integral forms from the divergence and Stokes theorems

Integrating the divergence equations over a volume V gives

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (4.1)$$

$$\oint_{\partial V} \mathbf{B} \cdot d\mathbf{A} = 0. \quad (4.2)$$

Integrating the curl equations over an oriented surface S gives

$$\oint_{\partial S} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{A}, \quad (4.3)$$

$$\oint_{\partial S} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{A}. \quad (4.4)$$

The integral forms expose global flux and circulation balances; the differential forms describe their local density.

5 Why displacement current is required

Taking the divergence of Ampere-Maxwell law and using $\nabla \cdot (\nabla \times \mathbf{B}) = 0$ yields

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}). \quad (5.1)$$

Gauss's law then gives

$$\frac{\partial \rho_q}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (5.2)$$

Without the displacement-current term, a time-varying charge density would contradict local charge conservation. Maxwell's correction is therefore not an optional decoration.

6 Electromagnetic waves in vacuum

Set $\rho_q = 0$ and $\mathbf{J} = 0$. Taking the curl of Faraday's law gives

$$\nabla \times \nabla \times \mathbf{E} = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}). \quad (6.1)$$

Using $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ and $\nabla \cdot \mathbf{E} = 0$,

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0. \quad (6.2)$$

The wave speed is

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}. \quad (6.3)$$

An identical equation follows for $\mathbf{B}$. Plane waves are transverse and satisfy $B = E/c$.

7 Potentials and gauge freedom

The identities

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} \quad (7.1)$$

automatically satisfy the two homogeneous Maxwell equations. The transformation

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla \chi, \quad \phi \rightarrow \phi - \frac{\partial \chi}{\partial t} \quad (7.2)$$

leaves $\mathbf{E}$ and $\mathbf{B}$ unchanged. Gauge freedom changes the mathematical representation, not measurable fields.

8 Low-frequency plasma limits

MHD typically assumes characteristic speeds much smaller than c and frequencies low enough that

$$\epsilon_0 \left| \frac{\partial \mathbf{E}}{\partial t} \right| \ll |\mathbf{J}|. \quad (8.1)$$

Ampere-Maxwell law then becomes approximately

$$\nabla \times \mathbf{B} \simeq \mu_0 \mathbf{J}. \quad (8.2)$$

Quasineutrality further assumes $|\rho_q| \ll en$, but it does not imply $\mathbf{E} = 0$. These approximations remove light waves and plasma-frequency charge oscillations from the resolved dynamics. Their domains of validity must be stated.

9 Constraint preservation in numerical models

The divergence equations act as constraints on admissible initial data. Faraday's law preserves $\nabla \cdot \mathbf{B} = 0$ analytically because

$$\frac{\partial}{\partial t}(\nabla \cdot \mathbf{B}) = -\nabla \cdot (\nabla \times \mathbf{E}) = 0. \quad (9.1)$$

A discrete scheme may violate the identity. Constrained transport, divergence cleaning, or compatible finite elements are therefore used to control numerical magnetic monopoles.

10 Computational laboratory

Discretize a one-dimensional vacuum electromagnetic wave and verify $c = 1/\sqrt{\mu_0\epsilon_0}$. Independently evaluate differential and integral forms of Gauss's law for a smooth charge distribution. Measure discrete charge-conservation and divergence errors.

11 Summary

- Maxwell's equations couple charges, currents, and time-dependent fields.
- The displacement current guarantees compatibility with charge conservation.
- Vacuum electromagnetic waves follow directly from the curl equations.
- MHD is a controlled low-frequency reduction, not a replacement for Maxwell theory.
- Divergence constraints require deliberate preservation in numerical simulations.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
ρ_q	charge density	Source in Gauss's electric law.
$\mathbf{J}$	current density	Charge flow per area.
c	speed of light	$1/\sqrt{\mu_0\epsilon_0}$.
$\mathbf{A}$	vector potential	Satisfies $\mathbf{B} = \nabla \times \mathbf{A}$.
χ	gauge function	Generates equivalent potentials.

Symbol / acronym	Name	Meaning in this document
$Q_{\text{enc}}, I_{\text{enc}}$	enclosed source mea- sures	Volume charge and surface-crossing current.

12 Exercises

1. Derive charge conservation from Ampere-Maxwell and Gauss laws.
2. Derive the vacuum magnetic-field wave equation.
3. Show that a plane electromagnetic wave is transverse.
4. Explain why quasineutrality does not eliminate electric fields.
5. Estimate when displacement current is negligible relative to conduction current.

13 References

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